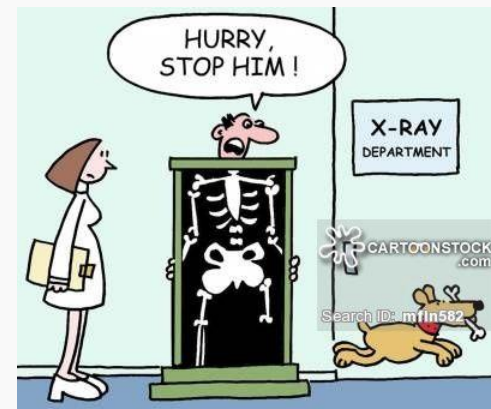
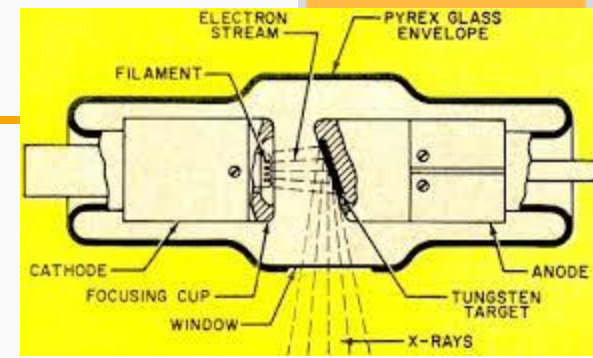


- Understand basic principles of x-rays
- Understand basic normal chest radiographic anatomy
- Formulate an approach to evaluating and reading chest radiographs





- An x-ray beam is generated by passing an electron beam through a vacuum between a cathode (-) and an anode (+).

- The positively charged anode attracts the rapidly moving, negatively charged electrons.
- As the electrons collide and interact with the atoms on the anode target, a great amount of energy is produced. 1% of this energy is in the form of x-radiation. 99% appears as heat and must be removed from the anode.
- The tube enclosure is shielded and a series of lead shutters allow the diagnostic beam to exit.



- Scientific unit of measurement for whole body radiation dose, called "effective dose," is the **millisievert (mSv)**.
- We use "**effective dose**" when they talk about the risk of radiation to the entire body. Risk refers to Radiation Dose in X-Ray and CT Exams refers to possible side effects, such as the chance of developing a cancer later in life.
- Effective dose takes into account how sensitive different tissues are to radiation. If you have an x-ray exam that includes tissues or organs that are more sensitive to radiation, your effective dose will be higher.

Darkest

Minimum x-ray

**Transmission
(least dense
tissue)**

air

fat

soft tissue/fluid*

calcium

bone

x-ray contrast

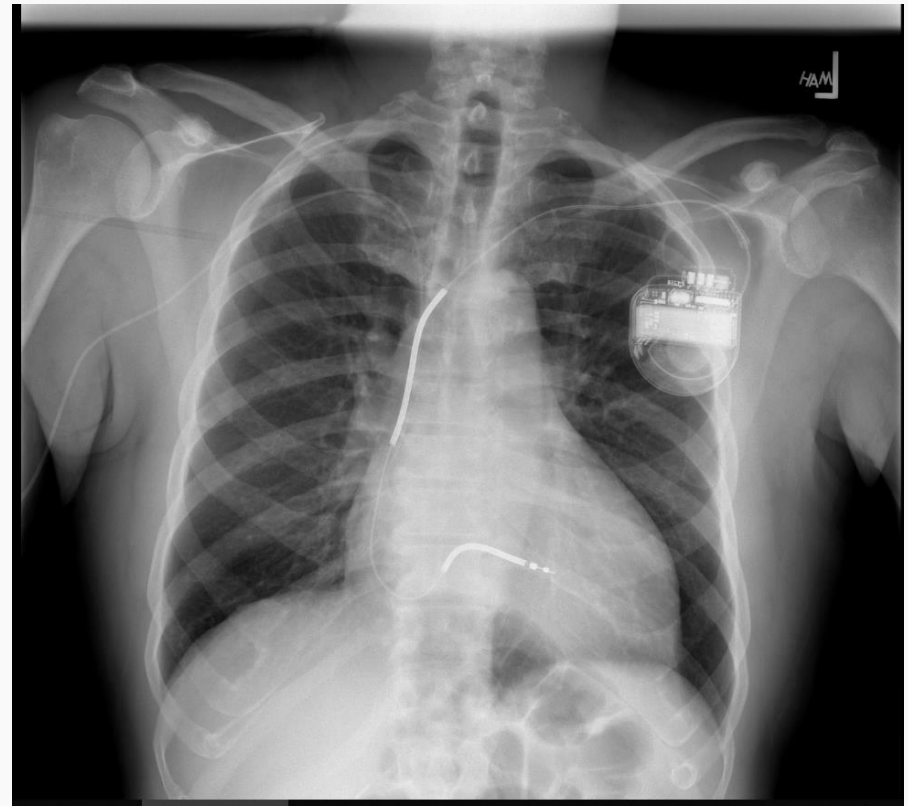
metal

Whitest

**Maximum x-ray
Absorption**

(densest tissue)

*fluid can be pus, blood or
simple fluid



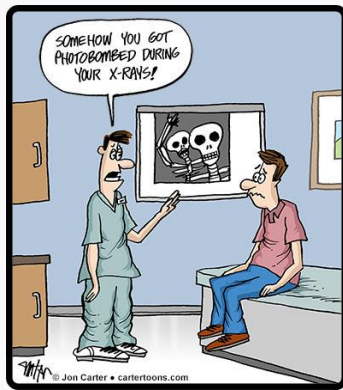


HAM

STITUTE

nic

- Basic Physics/ Radiation principles
- All of that physics basically creates the x-ray result which is essentially a shadow of the tissues that absorbed the radiation



- Understanding the radiograph

Tissue

contrast
(i.e their
densities)

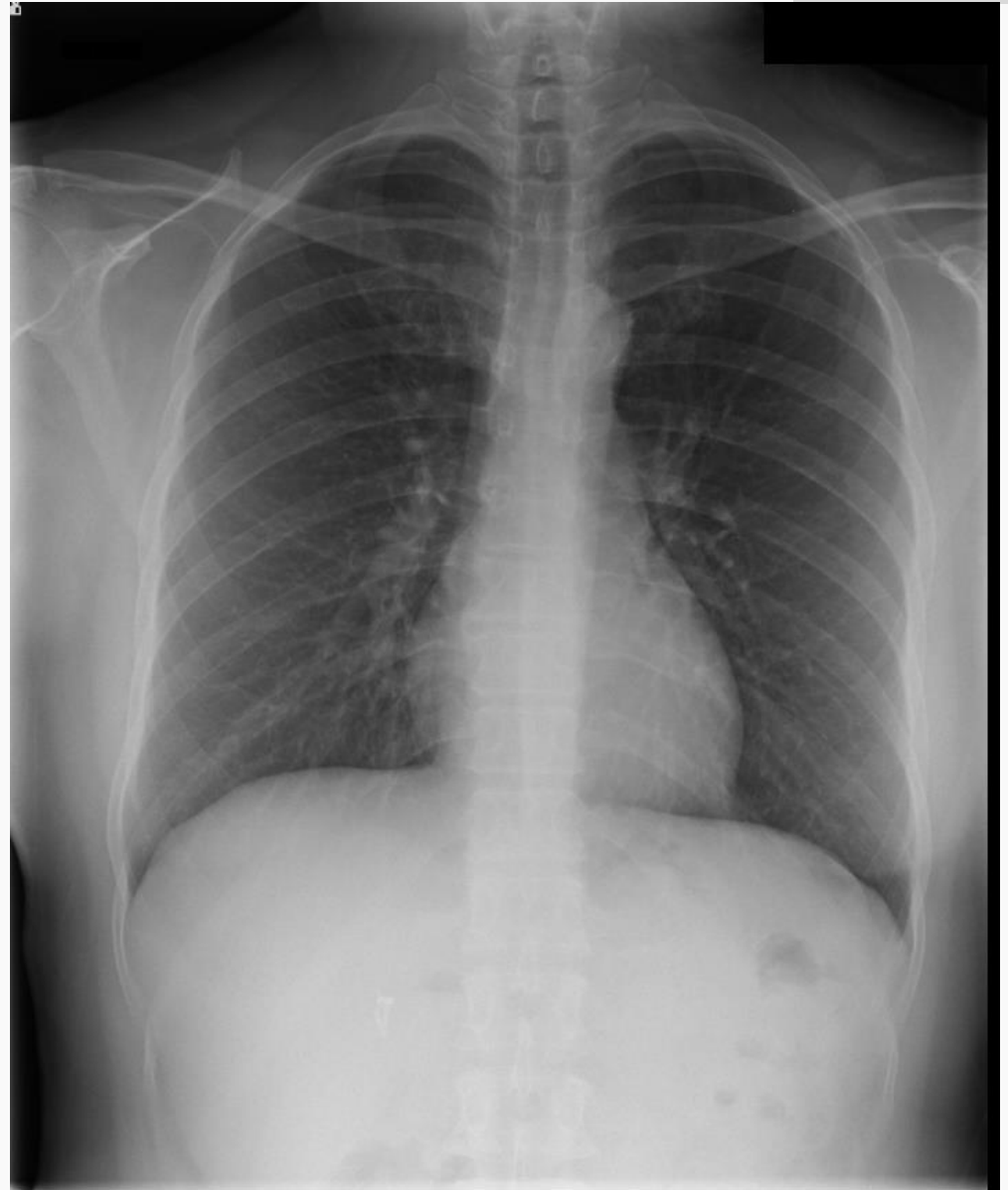


Image
contrast

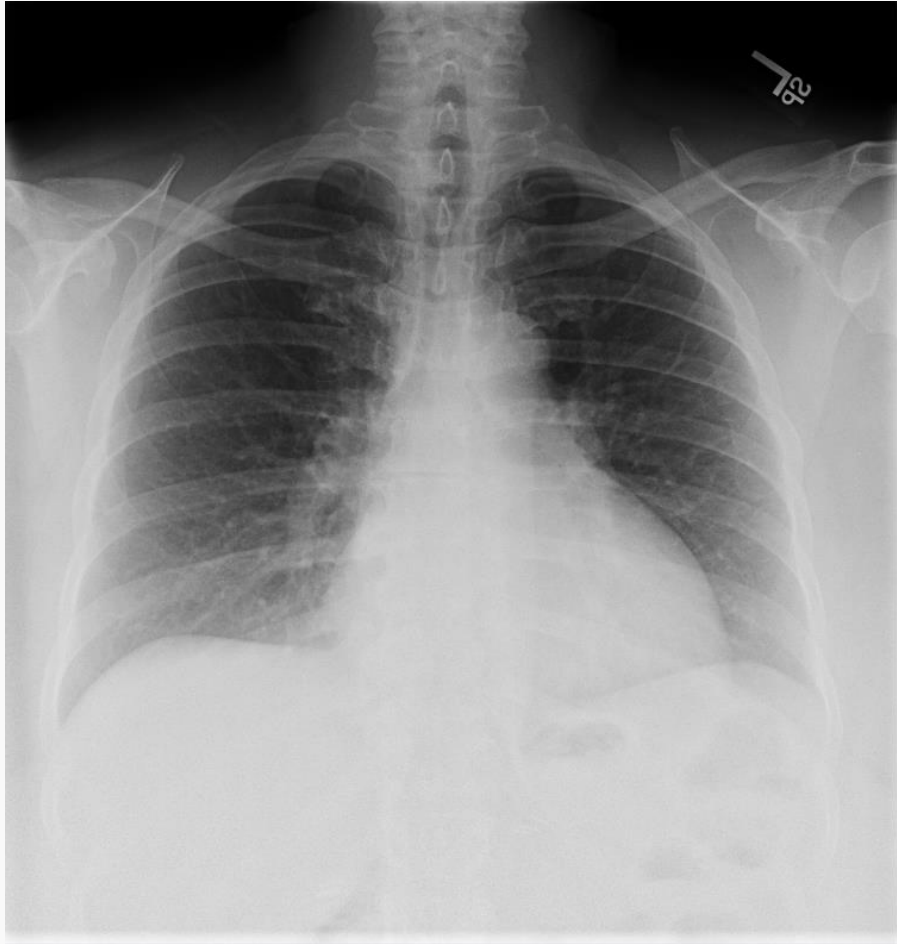
- Allows sharp margins between structures
- If not enough contrast (diff. in density) between adjacent tissues, no sharp margins between structures on the image and the objects will blend together- **Silhouette sign****

• Basic Physics/ Radiation principles

- A structure is *only* rendered visible on a radiograph by the juxtaposition of two different densities.
- When objects of the same density are next to each other they will be silhouetted
- For example: liver and diaphragm



**Silhouette Sign



Loss of the expected interface normally
created by juxtaposition of two structures of
different density
No boundary can be seen between two
structures of similar density

- Projection – PA or AP
 - How to tell → medial border of scapula (rule of thumb but not absolute)
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process



PA



AP

- Projection- Same patient

- Approach to the CXR: Technical Aspects

- Projection – PA or AP
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process



PA



AP

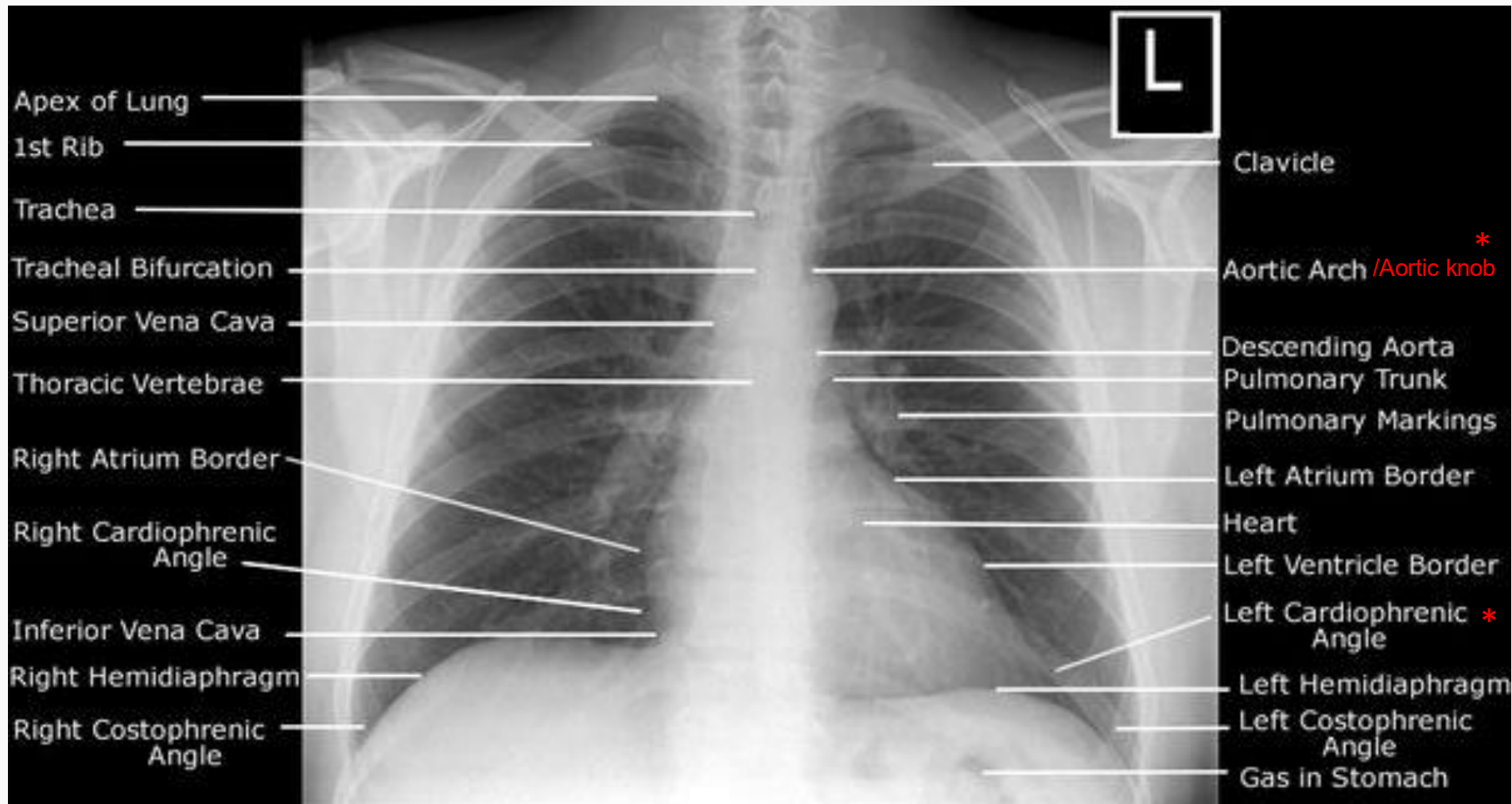
- Low lung volumes- Same patient

- Approach to the CXR: Technical Aspects

- Projection – PA or AP
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process

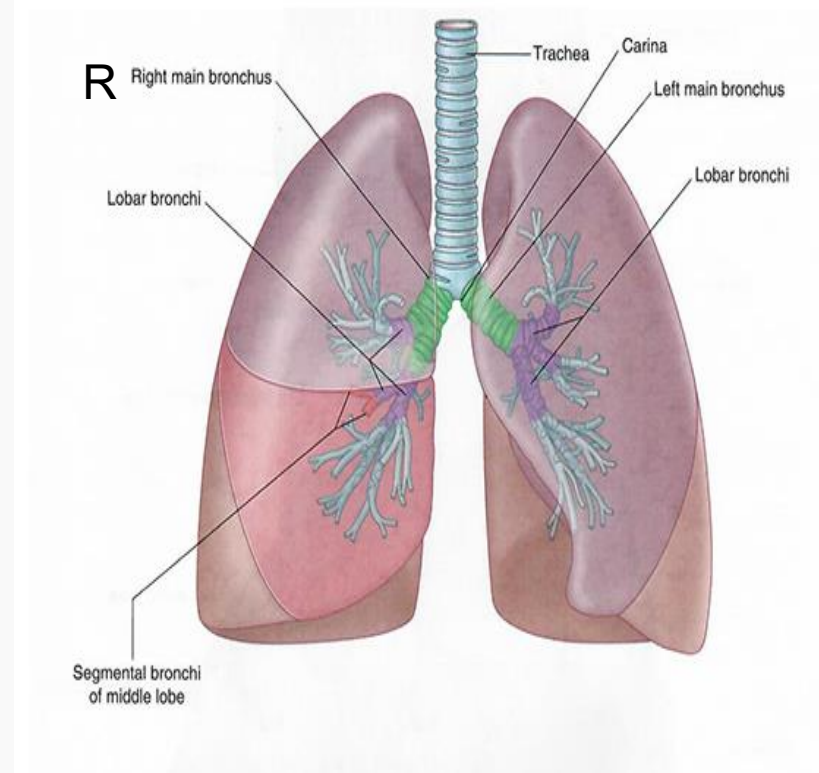
- Approach to the CXR: Technical Aspects
- Projection – PA or AP
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process

• Normal Frontal Chest X-ray



- Left heart border = left ventricle
- Right heart border= right atrium

- Fissures
 - Major fissure(oblique)-
right and left lungs
 - Minor fissure(horizontal)-
right



- Normal Anatomy

- The Key: A systematic approach

My approach:

- ▶ Assess study technique and quality
- ▶ Tubes and lines/Hardware/devices**
- ▶ Look at the partially imaged abdomen
- ▶ Bones and soft tissues
- ▶ Heart/Mediastinum
- ▶ Lungs and pleura- each one individually and then together for symmetry
 - ▶ OR
- ▶ **ABCDE method**

